ACIDS, BASES AND SALTS

COURSE OUTLINE

- Introduction
- Acids and Bases
  - Definition
  - Physical and Chemical properties
  - Uses
- Identification and Measurement
  - Indicators and pH scale
  - Importance of pH
- Salts
  - Definition
  - Properties and uses

INTRODUCTION

- **Acids, bases** and **salts** are 3 distinctive classes of chemical compounds
- Important part of chemistry
- All substances are **acidic**, **neutral** or **basic** (alkaline)
- 3 groups of compounds are found everywhere
  - Foods, medicines, cleaning products

ACIDS

Derived from latin word *acidus* meaning sour

- Two common definitions of an acid
  - Arrhenius theory
  - Bronsted-Lowry theory

- In 1890, Arrhenius **DEFINED** acids as:

  A substance that produces hydrogen ions (H+) when dissolved in aqueous solution



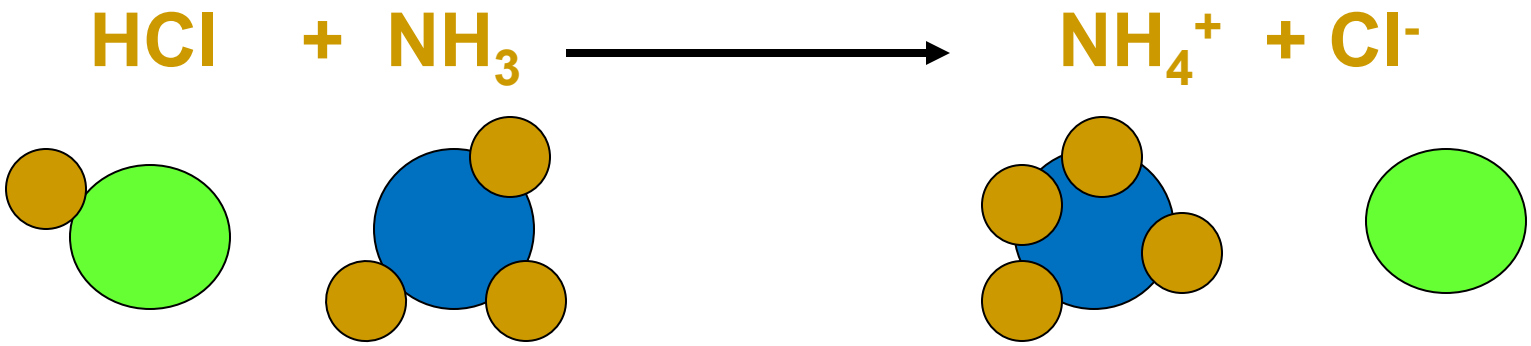
ACIDS

- Arrhenius definition useful for describing acids in certain reactions
- Limited in its scope
  - Acid must be soluble in water
  - Requires base to have a OH^-
- What if acid does not produce H^+ in a reaction or base does not have OH^- group?

ACIDS

- In 1923, Bronsted-Lowry proposed a more general theory
- **DEFINED** an acid as:

  a substance that donates a proton or hydrogen ion (H+)



Physical properties of acids

- Acids have a sour taste e.g. vinegar
- Acids are corrosive
- Acids have a pH less than 7
- Turn blue litmus paper red
- Turn universal indicator orange or red
- Conduct electricity

Chemical properties of acids

- Acids react with bases to give salt and water i.e. neutralization

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

- Acids react with metals to give hydrogen gas

$$Zn + H_2SO_4 \longrightarrow H_2 + ZnSO_4$$

- Acids react with carbonates to give carbon dioxide

$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$

- Release H+ ions in water solutions

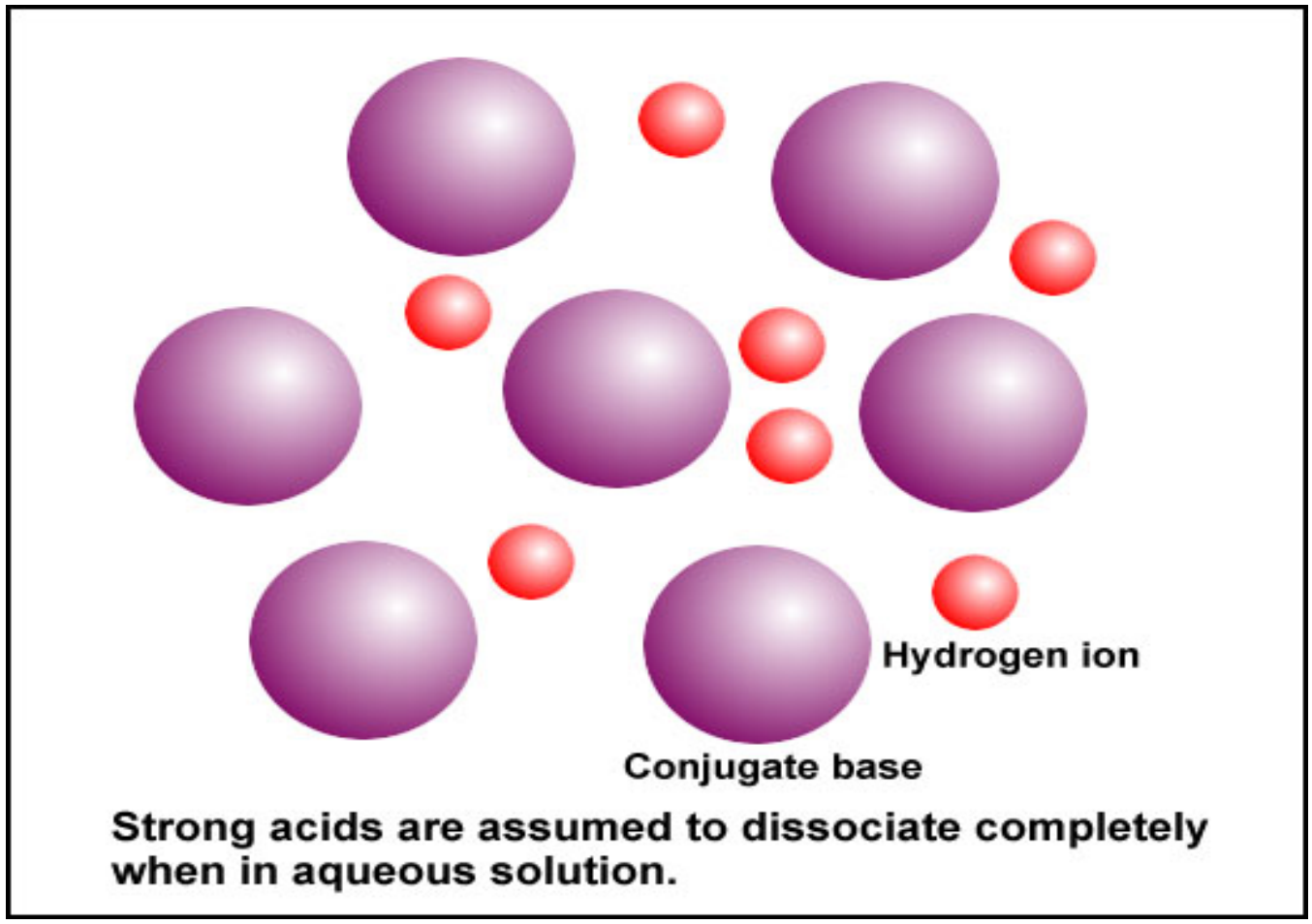
$$HCl\ (aq) \longrightarrow H^+\ (aq) + Cl^-\ (aq)$$

Acidity and Water

- A dilute acid has lots of water and a small amount of acid
- A concentrated acid has lots of acid and not much water so must be handled carefully
- A strong acid releases lots of H^+ (ionizes completely in water)
- A weak acid releases fewer H^+ (partially ionizes in water)

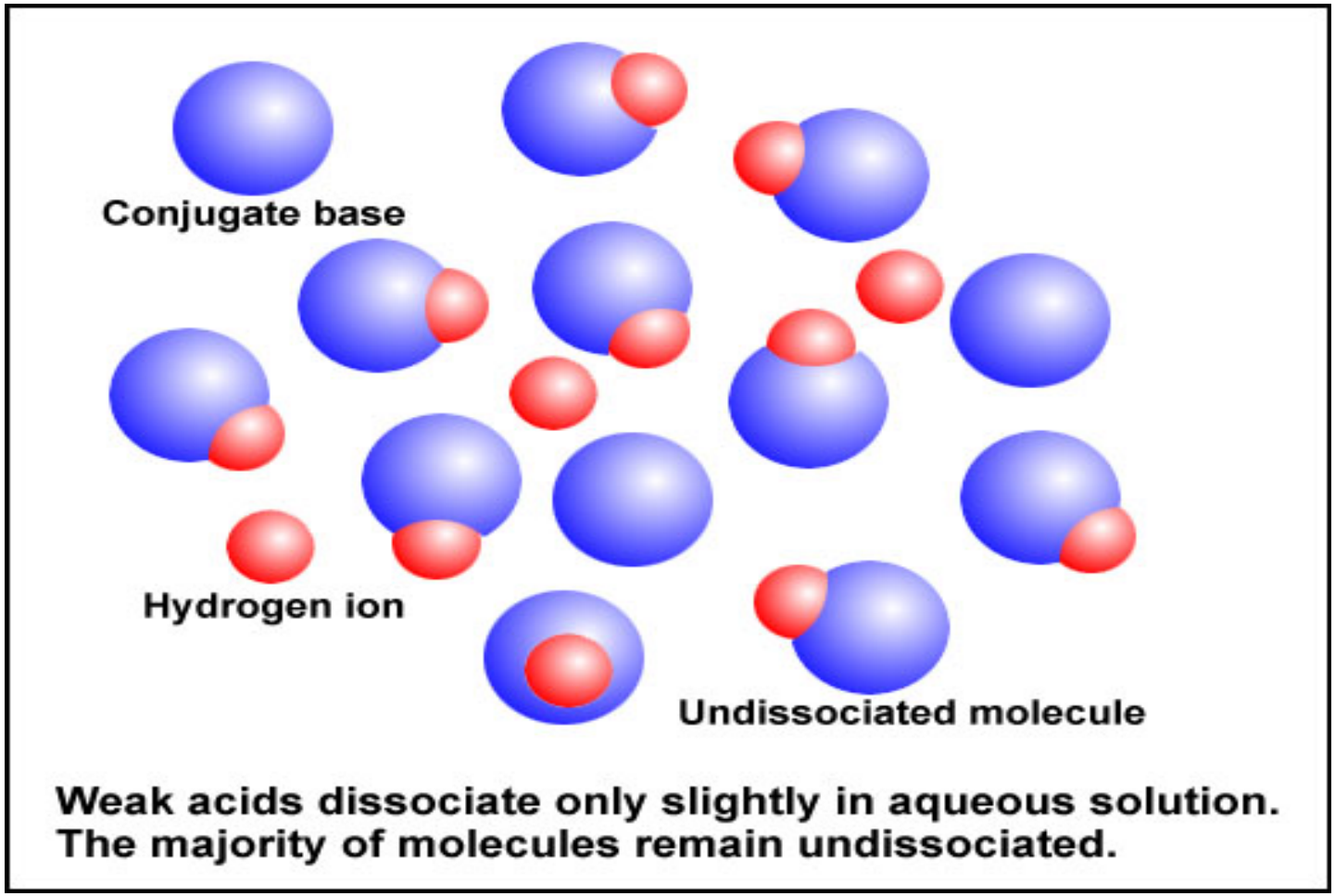
Acidity and Water

- Strong acid



Acidity and Water

- Weak acid



Acid Reactions

- Acid + Base → Salt + Water
- Acid + Metal → Salt + Hydrogen
- Acid + Carbonate → Salt + Water + Carbon Dioxide
  - Hydrochloric acids (HCl) form CHLORIDE salts
  - Nitric acid (HNO_3) forms NITRATE salts
  - Sulfuric acid (H_2SO_4) forms SULFATE salts

Some common acids

HCl	hydrochloric acid
HNO_3	nitric acid
H_3PO_4	phosphoric acid
H_2SO_4	sulfuric acid
CH_3COOH	acetic acid (Vinegar)

Uses of acids

- Sulphuric acid:
  - Manufacture of non-soapy detergents
  - Electrolyte in lead-car batteries
  - Removal of oxides from iron and steel before galvanizing and electroplating
- Nitric acid:
  - Manufacture of fertilizers, plastics, explosives

Uses of acids

- Hydrochloric acid:
  - aids in digestion in stomach
- Acetylsalicylic acid (aspirin):
  - Used as an analgesic, treatment of headache and fever
- Ascorbic acid (vitamin C):
  - Used to prevent and treat scurvy
  - Found in citrus fruits e.g. lemon

BASES

- Chemically opposite of acids
- Base **DEFINED** as:

  a substance that can accept hydrogen ions (protons) and neutralize an acid
- The reaction between an acid and a base is called **NEUTRALIZATION** (7pH)
- Neutralization always forms a salt and water

ACID + BASE ⟶ SALT + WATER

BASES

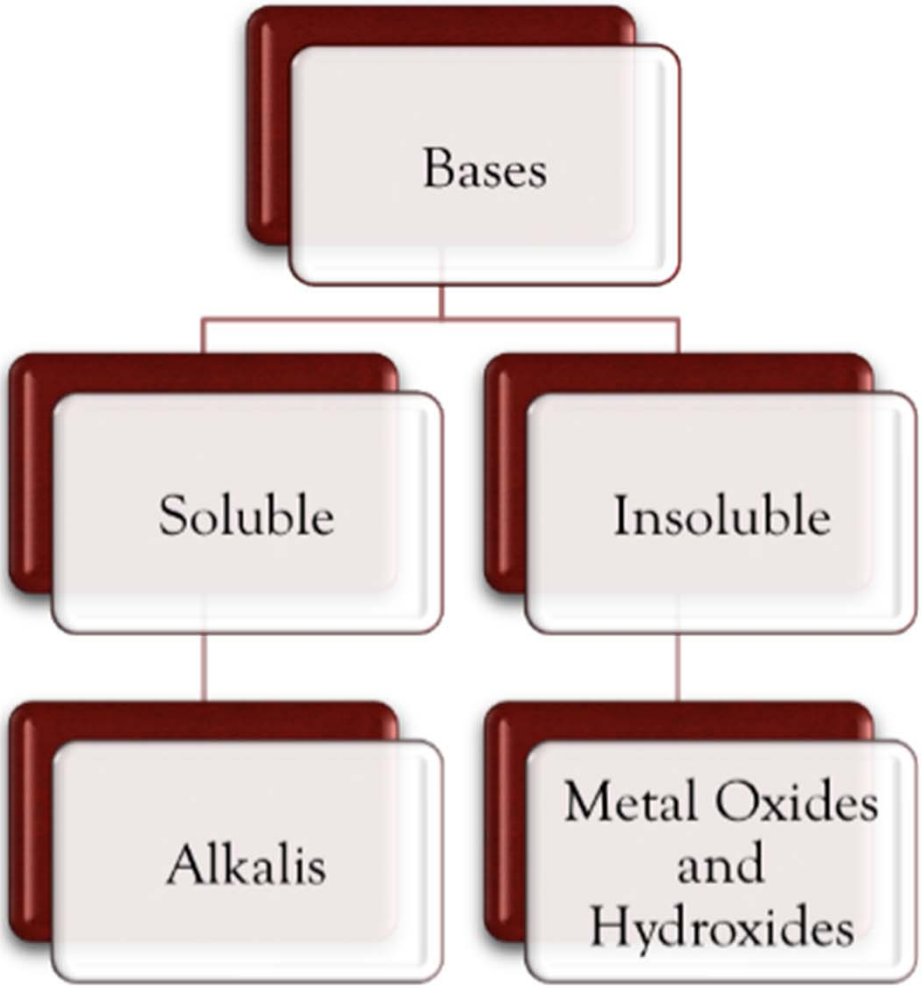
- Bases are compounds of:
  - Metal oxides e.g. Na_2O
  - metal hydroxides e.g. NaOH
  - metal carbonates e.g. Na_2CO_3
  - metal hydrogen carbonates e.g. $NaHCO_3$
  - ammonia solution e.g. NH_3
- a water soluble base is referred to as an **ALKALI**
- Alkalis dissociate in water to give hydroxyl ions (OH^-)

Properties of bases

- Feels slippery or soapy
- Tastes bitter
- pH of bases greater than 7
- Turns red litmus paper blue
- Produce OH- in water



Family of bases



Family of bases

Insoluble bases	Soluble bases
Copper (II) oxide (CuO)	Sodium hydroxide (NaOH)
Iron (III) oxide (Fe_2O_3)	Potassium hydroxide (KOH)
Copper (II) hydroxide ($Cu(OH)_2$)	Calcium hydroxide ($Ca(OH)_2$)
Lead (II) oxide (PbO)	Ammonia solution ($NH_{3(aq)}$)

Some common bases

- Calcium hydroxide $Ca(OH)_2$
- Sodium hydroxide NaOH
- Sodium hydrogen carbonate $NaHCO_3$
- Magnesium hydroxide $Mg(OH)_2$
- Ammonia NH_3

Uses of bases

- Ammonia:
  - Used as a cleaning agent
  - Preparation of nitric acid and ammonium chloride
- Sodium hydroxide:
  - Used to make soap and detergents

Uses of bases

- Calcium oxide
  - Used to make iron, concrete and cement
- Magnesium hydroxide:
  - Used in toothpaste to neutralize acid
  - Used in antacids to relieve indigestion

Indicators

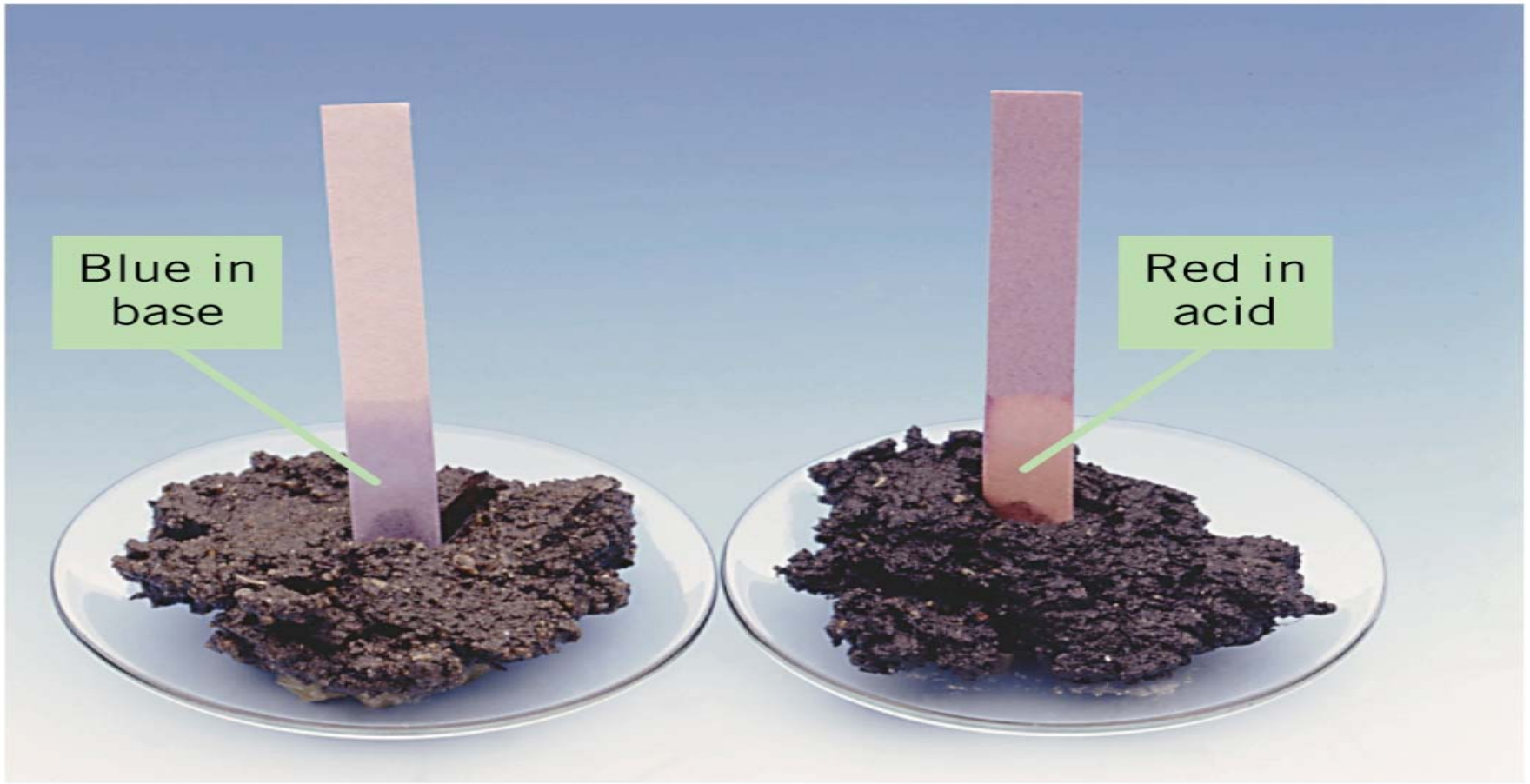
- To identify whether a substance or solution is acidic or basic, indicators are used
- Indicators **DEFINED** as:

  substances that change colour in the presence of an acid or base

- two common indicators:
  - Litmus paper and Universal indicator

Indicators

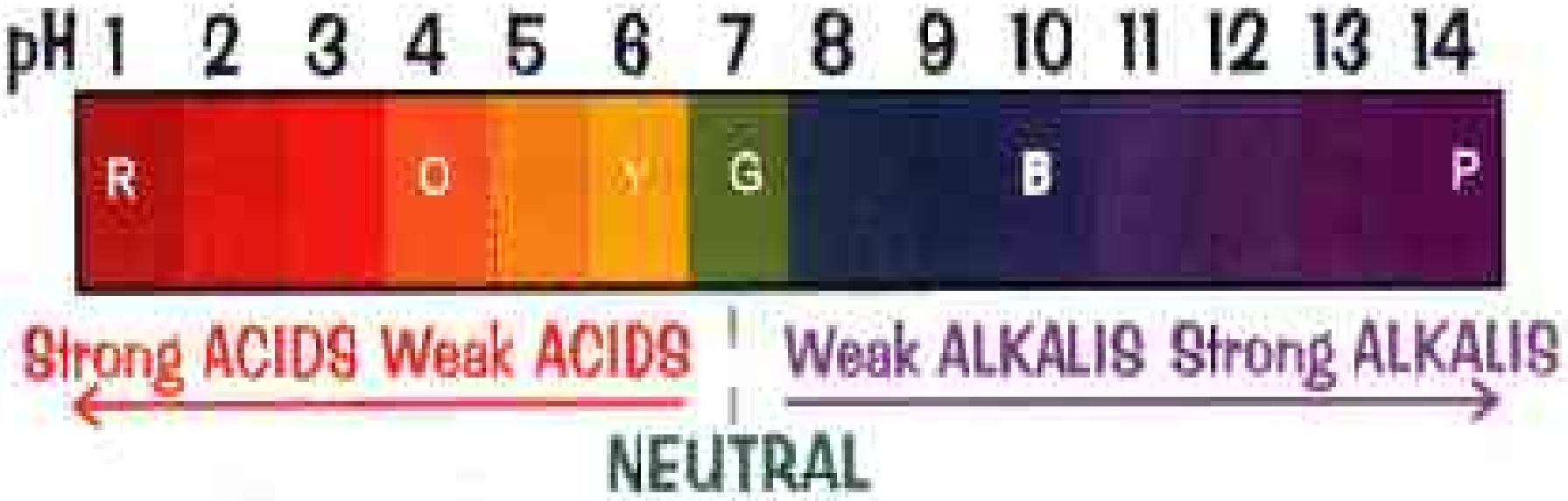
- **RED** litmus paper turns **BLUE** in a basic media
- **BLUE** litmus paper turns **RED** in an acid media



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Indicators

- Universal indicator shows series of colour changes
- Shows different strengths of acids and bases
  - The stronger the acid, the lower the pH
  - The stronger a base, the higher the pH



Indicators

Indicator	Colour in acids	Colour in alkalis
Phenolphthalein	Colourless	Pink
Methyl orange	Pink	Yellow
Methyl red	Red	Yellow
Bromothymol blue	Yellow	Blue

pH

- pH stands for potential Hydrogen
- Measures the acidity and alkalinity of a solution
- pH value is directly proportional to [H+] ions (for acids) or [OH-] ions (for alkalis), pH scale ranges from 0 to 14
- pH can be measured using a) pH meter and b) pH indicator paper

pH

- For neutral solutions, pH = 7
- For acidic solutions, pH < 7
- For basic solutions, pH > 7

Importance of pH

- Living organisms i.e. humans, plants, animals are extremely sensitive to pH
- Function best at certain narrow pH range
- In agriculture, the pH of the soil is monitored as certain crops grow better in a certain pH range
  - rice = slightly acidic
  - Citrus fruits = alkaline
- In medicine, certain disease are diagnosed by the pH of urine and blood

Salts

- a salt is **DEFINED** as:

  An ionic compound which is formed by neutralization reaction between an acid and a base

- **Ionic = Cation (+ve) + Anion (-ve)**
- Some salts are soluble in H_2O, some are insoluble

Common Salts

Chemical name	Common name	Formula
Calcium oxychloride	Bleaching powder	$CaOCl_2$
Sodium carbonate decahydrate	Washing soda	$Na_2CO_3.10H_2O$
Sodium hydrogen carbonate	Baking soda	$NaHCO_3$
Sodium hydroxide	Caustic soda	NaOH
Sodium chloride	Common salt	NaCl

Uses of salts

- **Calcium oxychloride** (Bleaching powder)
  - Disinfection of water and swimming pools
  - Bleaching cotton and linen
  - Used as an oxidising agent in many chemical industries
  - Ingredient in bathroom cleaners and disinfectant sprays
  - Manufacture of chloroform

Uses of salts

- **Sodium carbonate decahydrate (Washing soda)**
  - Manufacture of sodium compounds e.g. borax, caustic soda
  - Used in glass, soap and paper manufacturing industries
  - Used for softening hard water
  - Used as a cleaning agent and as a lab reagent

Uses of salts

- **Sodium hydroxide (Caustic soda)**
  - Used in drain and oven cleaners
  - Used as a bleaching agent for fabrics
  - Used in the refinery of bauxite to produce aluminium oxide

Uses of salts

- **Sodium hydrogen carbonate (Baking soda)**
  - Ingredient in antacids
  - Additive in food and drinks
  - Used to make baking powder
  - Used in soda-acid fire extinguishers

Salts

- Salts are either soluble or insoluble in H_2O

Soluble	Insoluble
All Nitrates	-
All sulphates except -->	$BaSO_4$, $CaSO_4$, $PbSO_4$
All Chlorides except -->	$PbCl_2$ (soluble in hot water), AgCl, $HgCl_2$
Potassium, Sodium, Ammonium salts	-
K_2CO_3, Na_2CO_3, $(NH_4)_2CO_3$	All other carbonates
K_2O, Na_2O	All other oxides

THE END